

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration, Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks:50

Annexure A SENARIO BASED QUESTIONS

<i>Criteria</i>	Understanding of Scenario (2)	Identification of Key Issues (2)	Application of Concepts (2.5)	Decision Making (2)	Clarity of Response (1.5)	<i>Total(10)</i>
<i>Weightage</i>	20%	20%	25%	20%	15%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I N.Tejaswitha 192371057 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

N.Tejaswitha

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A
Scenario based Questions

Q1. Data Preprocessing in Retail Analytics (L3, L4)

A **retail analytics company** is preparing customer purchase data for building a recommendation system. The dataset contains missing values, inconsistent categorical formats, and extreme values due to data collected from multiple stores and manual entries.

Sample Dataset

Customer_ID	Age	Gender	Monthly_Spend (₹K)	Purchase_Frequency	Loyalty_Level
C1	23	M	20	5	Low
C2	NaN	Male	35	8	Medium
C3	31	F	NaN	10	High
C4	105	Female	500	50	High
C5	27	male	25	NaN	Low
C6	40	FEMALE	45	12	Medium
C7	36	M	38	9	Low
C8	42	Female	300	45	High

Tasks

A. Handling Missing Values

- Identify missing values in **Age**, **Monthly_Spend**, and **Purchase_Frequency**
 - Apply at least **two imputation techniques** (e.g., Median Imputation, k-NN Imputation)
 - Compute the imputed values and justify the selected method
-

B. Outlier Detection and Treatment

- Detect outliers in **Age** and **Monthly_Spend** using the **Z-score method**
 - Identify records containing extreme values (e.g., Age = 105, Spend = 500)
 - Apply suitable treatment methods (**capping** / **transformation** / **removal**) and justify
-

C. Categorical Data Standardization

- Identify inconsistencies in **Gender** (M, Male, male, FEMALE, etc.)
- Convert all entries into a standardized format (**Male** / **Female**)

Q2. Covariance & Data Reduction in E-commerce Analytics (L4, L5)

An **e-commerce platform** is building a predictive model to classify high-value customers. The dataset includes several correlated financial and behavioral attributes, leading to redundancy and increased computational cost.

Sample Dataset

Cust_ID	Age	Income (₹K)	Purchase_Value (₹K)	Credit_Score	Orders	Avg_Balance (₹K)	EMI (₹)	Customer_Class
U1	24	40	100	650	15	12	5000	Low
U2	34	60	150	700	20	18	7000	Medium
U3	44	85	210	720	28	25	9000	Medium
U4	52	120	320	800	35	40	15000	High
U5	29	45	110	680	18	14	5500	Low
U6	39	75	180	710	24	22	8500	Medium
U7	56	150	360	820	40	50	18000	High
U8	37	70	160	705	22	20	8000	Medium

Tasks

A. Covariance Analysis

- i. Compute covariance between **Income and Purchase_Value** (show steps or partial calculation)
- ii. Interpret whether the relationship is **positive or negative**
- iii. Explain how covariance helps in identifying feature relationships

B. Feature Selection

- i. Identify redundant attributes using **correlation or domain knowledge**
- ii. Select an optimal subset of features for modeling
- iii. Justify your selection

C. Data Transformation

- i. Standardize the dataset (excluding **Cust_ID** and **Customer_Class**)
- ii. Explain why normalization/standardization is required

Q3. Data Transformation & Discretization in Education Analytics (*L3, L4, L5*)

An **education analytics team** is developing a model to predict student performance. Continuous variables such as study hours and test scores need to be discretized to improve interpretability and decision-making.

Sample Dataset

Student_ID	Study_Hours	Test_Score	Attendance (%)	Performance
S1	2	40	60	Poor
S2	3	50	65	Average
S3	4	55	70	Average
S4	5	65	75	Good
S5	6	75	80	Good
S6	7	85	85	Good
S7	8	90	88	Excellent
S8	9	95	90	Excellent
S9	10	98	92	Excellent
S10	11	100	95	Excellent

Tasks

A. Equal-Width Binning

- Apply **3 bins** for Study_Hours and Test_Score
 - Calculate bin width and define intervals
-

B. Equal-Frequency Binning

- Divide data into **3 bins with equal records**
 - Assign values accordingly
-

C. Concept Hierarchy Construction

- Study_Hours → Low / Medium / High

- Test_Score → Poor / Average / Excellent

4. Use the two methods below to normalize the following group of data:

200, 300, 400, 600, 1000

(a) min-max normalization by setting min = 0 and max = 1

(b) z-score normalization

5. Data Transformation in Online Shopping Behavior Analysis (L3, L4, L5)

An **e-commerce company** is analyzing customer engagement by measuring the **time spent (in seconds)** on product pages. The duration varies significantly—from a few seconds to more than an hour—resulting in highly skewed data.

To improve the effectiveness of **customer segmentation using clustering techniques**, the dataset must be properly transformed and normalized.

Sample Dataset: Visit Duration

Customer_ID Visit_Duration (seconds)

C1	15
C2	30
C3	45
C4	60
C5	120
C6	300
C7	600
C8	900
C9	1800
C10	3600

Tasks

A. Equal-Frequency Binning

- Divide the dataset into **3 bins** with approximately equal number of records
 - Assign each value to its respective bin
 - Analyze how effectively this method groups short and long visit durations
-

B. Z-Score Normalization

- i. Compute the **mean and standard deviation** of visit duration
- ii. Apply Z-score normalization:

$$Z = \frac{X - \mu}{\sigma} = -$$

- iii. Discuss how this method handles **extreme values (e.g., 1800, 3600 seconds)**
-

C. Min-Max Normalization

- i. Apply Min-Max normalization using range **[0,1]**:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

- ii. Transform all values accordingly
- iii. Compare results with Z-score normalization in terms of **scaling and sensitivity to outliers**

Q1 : Data Preprocessing in Retail Analytics

1. Introduction

The retail customer purchase dataset contains customer information such as Customer ID, Age, Gender, Monthly Spend, Purchase Frequency, and Loyalty Level. Since the data was collected from multiple stores and manual entries, it contains missing values, inconsistent categorical values, and extreme values. Data preprocessing is performed to improve the quality and reliability of the dataset before applying data mining techniques.

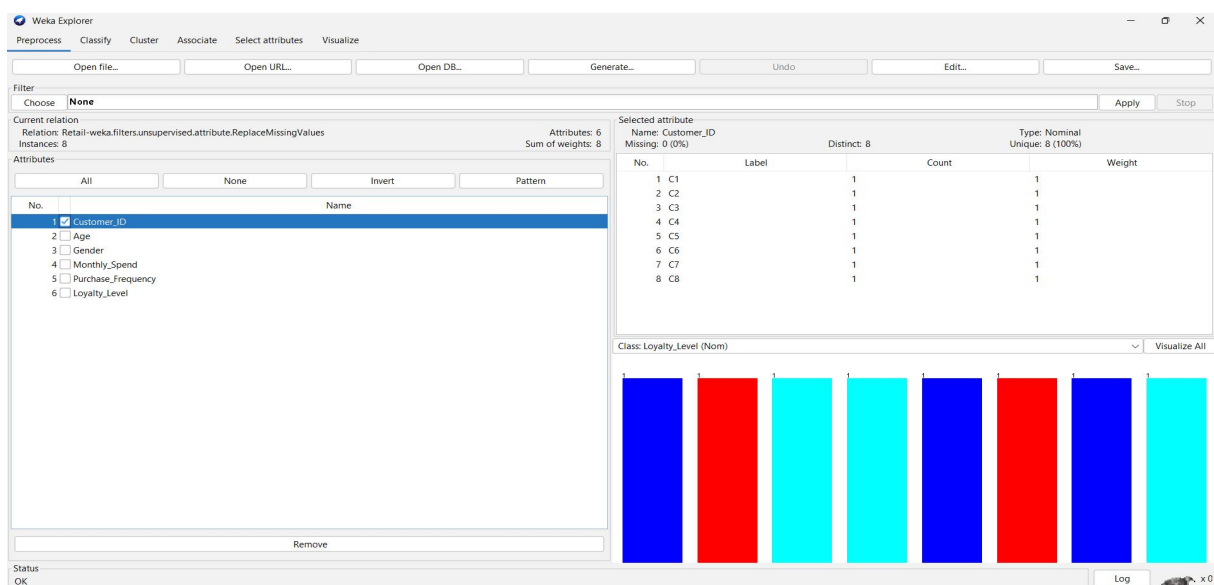
A. Handling Missing Values

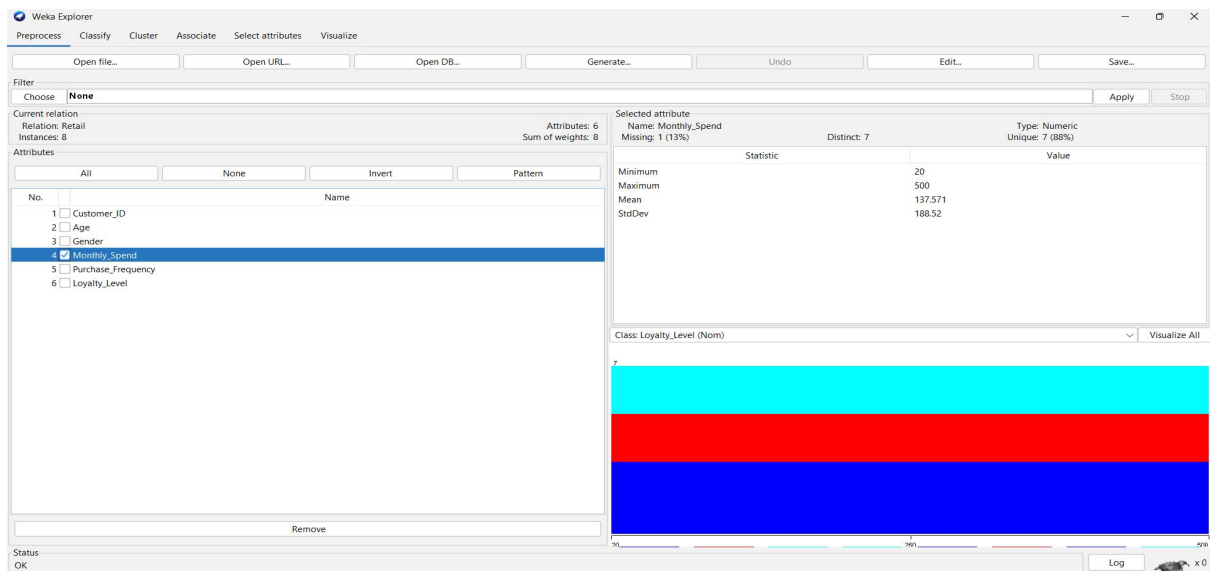
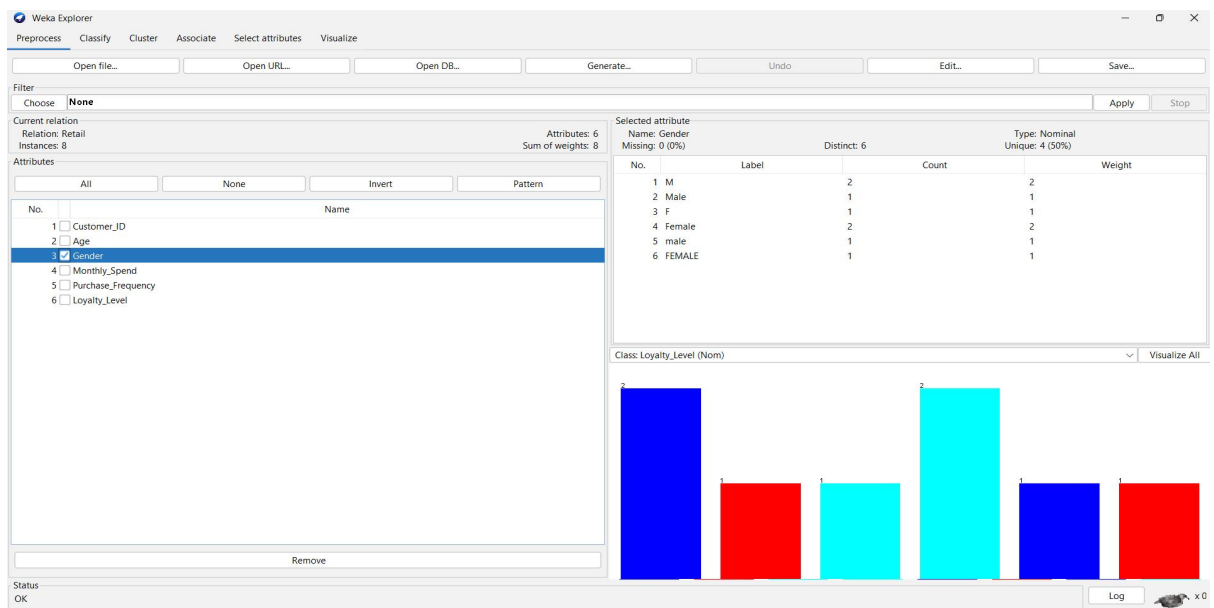
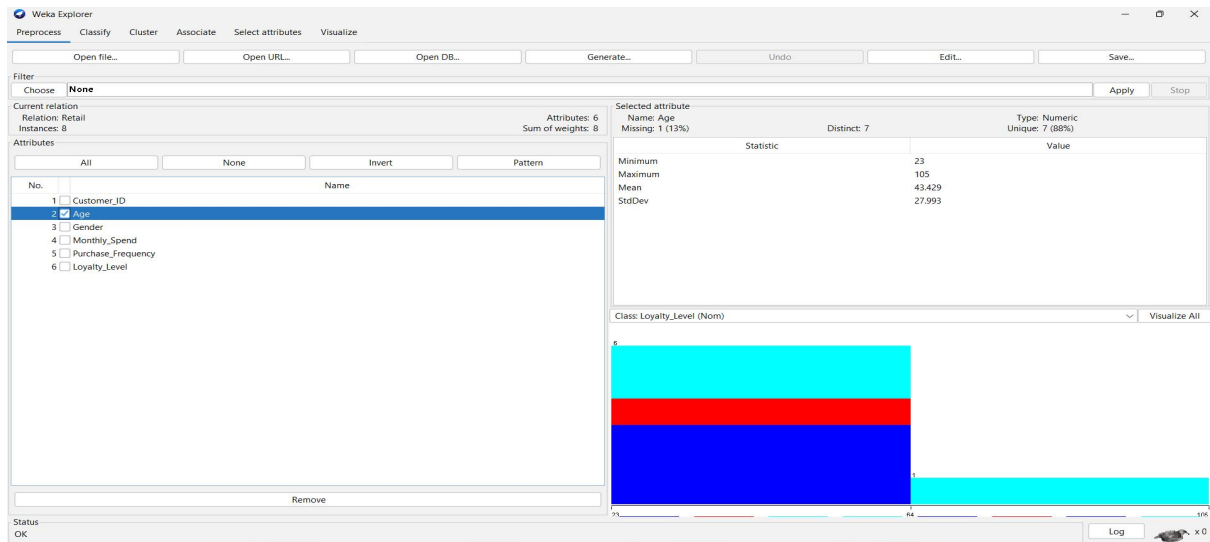
(i). Identification of Missing Values

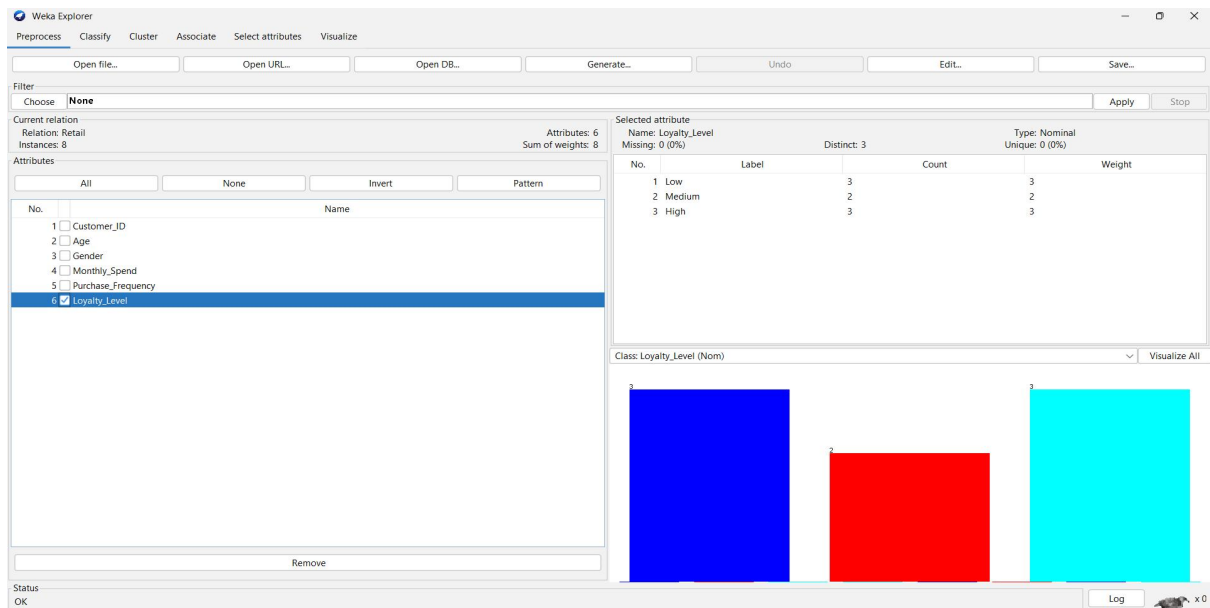
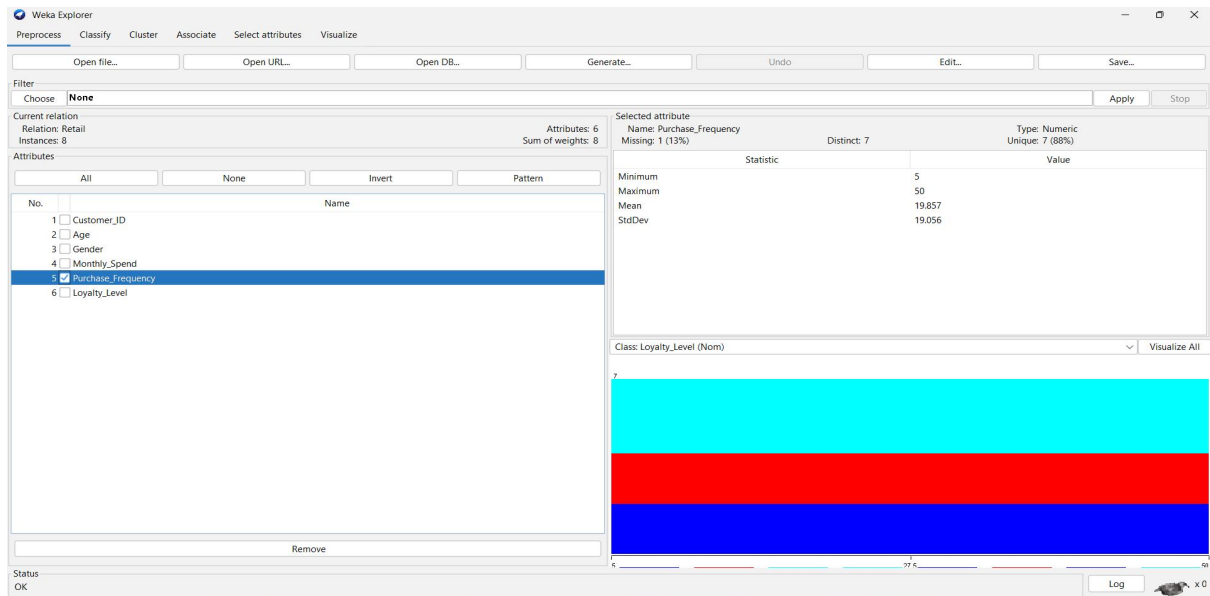
The dataset was loaded into WEKA and each attribute was examined. The following missing values were identified:

Attribute	Customer ID	Missing Value
Age	C2	?
Monthly_Spend	C3	?
Purchase_Frequency	C5	?

MISSING VALUES :







(ii). Applying Imputation Techniques

Two imputation techniques are commonly used to replace missing values:

Technique 1: Median Imputation

The missing numerical values are replaced with the median of the corresponding attribute.

This method is suitable because the dataset contains outliers, and the median is less affected by extreme values.

Technique 2: k-NN Imputation

The missing values are replaced using the values of the nearest neighboring records based on similarity.

This method preserves relationships among attributes and provides more accurate estimates for missing data.

In WEKA, the ReplaceMissingValues filter was applied to replace the missing values in the dataset.

REPLACING MISSING VALUES :

The screenshot shows the Weka Explorer interface with the 'Preprocess' tab selected. The 'Filter' dropdown is set to 'ReplaceMissingValues'. The 'Current relation' is 'Retail-weka.filters.unsupervised.attribute.ReplaceMissingValues' with 8 instances. The 'Attributes' list on the left shows 'Customer_ID' selected. The 'Selected attribute' panel on the right shows 'Name: Customer_ID', 'Missing: 0 (0%)', 'Distinct: 8', and 'Type: Nominal Unique: 8 (100%)'. Below this, a table lists the distinct values:

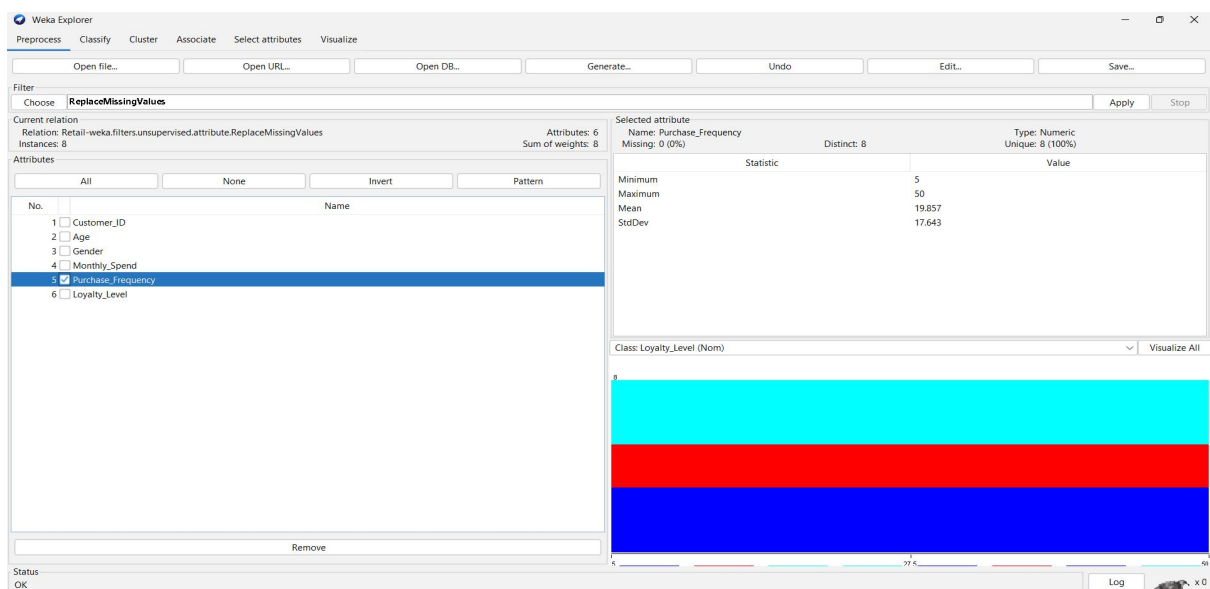
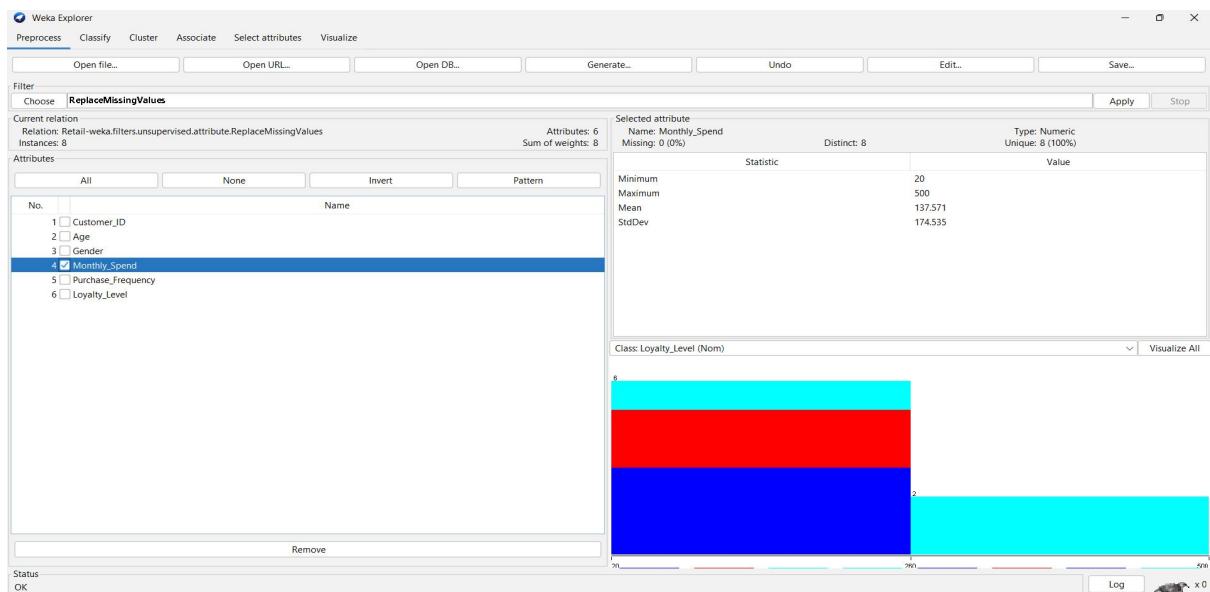
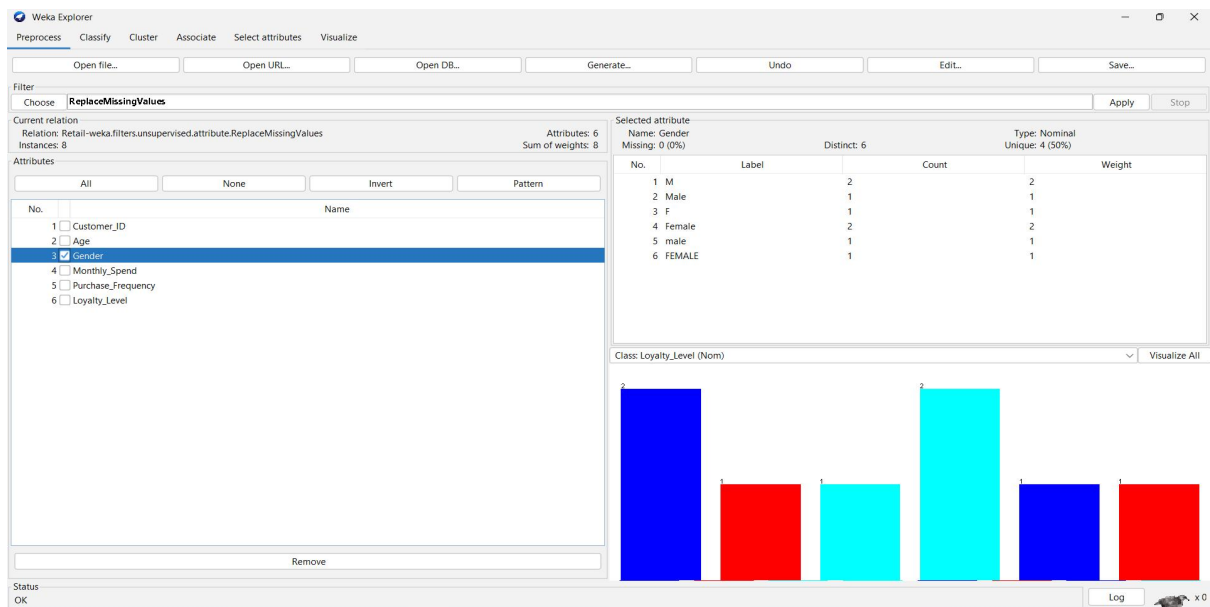
No.	Label	Count	Weight
1	C1	1	1
2	C2	1	1
3	C3	1	1
4	C4	1	1
5	C5	1	1
6	C6	1	1
7	C7	1	1
8	C8	1	1

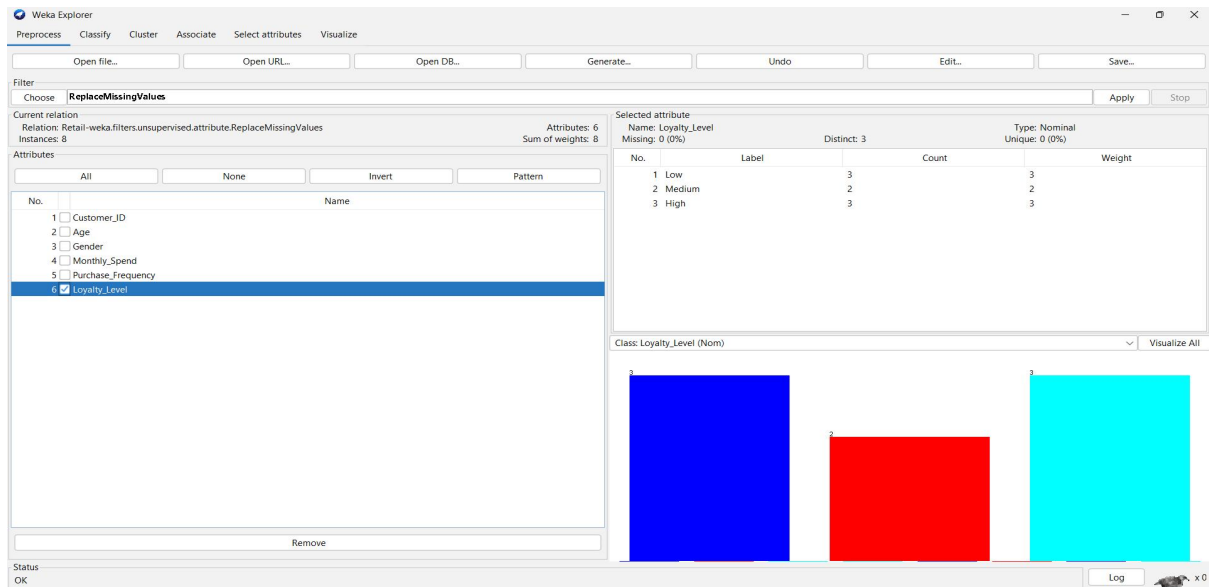
Below the table, a bar chart visualizes the distribution of the 'Loyalty_Level' (Nom) attribute, showing eight bars of varying heights and colors (blue, red, cyan).

The screenshot shows the Weka Explorer interface with the 'Preprocess' tab selected. The 'Filter' dropdown is set to 'ReplaceMissingValues'. The 'Current relation' is 'Retail-weka.filters.unsupervised.attribute.ReplaceMissingValues' with 8 instances. The 'Attributes' list on the left shows 'Age' selected. The 'Selected attribute' panel on the right shows 'Name: Age', 'Missing: 0 (0%)', 'Distinct: 8', and 'Type: Numeric Unique: 8 (100%)'. Below this, a table lists the statistics for the 'Age' attribute:

Statistic	Value
Minimum	23
Maximum	105
Mean	43.429
StdDev	25.917

Below the table, a bar chart visualizes the distribution of the 'Loyalty_Level' (Nom) attribute, showing three bars of varying heights and colors (cyan, red, blue).





(iii). Imputed Values and Justification

After applying the preprocessing technique, the missing values were successfully replaced. The completed dataset no longer contained missing values.

The selected preprocessing method improves data quality by preventing information loss and making the dataset suitable for further analysis. Median imputation is preferred for datasets containing outliers, while k-NN imputation provides more accurate estimates by considering similar records.

B. Outlier Detection and Treatment

(i). Detecting Outliers Using the Z-Score Method

The numerical attributes Age and Monthly_Spend were examined for unusually high values. The Z-score method helps identify observations that deviate significantly from the mean. Records with very high Z-score values are considered outliers.

(ii). Identification of Extreme Values

The following records contain extreme values:

Customer ID	Attribute	Value
C4	Age	105
C4	Monthly_Spend	500
C8	Monthly_Spend	300

These values are much higher than the remaining observations and are treated as outliers.

(iii). Outlier Treatment

Outliers can be handled using different techniques such as:

- **Capping** – Replace extreme values with an upper limit.
- **Transformation** – Apply logarithmic or square root transformation.
- **Removal** – Remove records if they are incorrect or invalid.

In this dataset, capping is an appropriate choice because it reduces the influence of extreme values while preserving all customer records for analysis.

C. Categorical Data Standardization

(i). Identification of Inconsistent Gender Values

The Gender attribute contains different representations of the same category.

Original Values

M
Male
male
F
Female
FEMALE

These inconsistent values may lead to duplicate categories during data analysis.

(ii). Standardizing the Gender Attribute

The Gender values were converted into a consistent format.

Original Value	Standardized Value
M	Male
Male	Male
male	Male
F	Female
Female	Female
FEMALE	Female

This standardization ensures that only two categories (**Male** and **Female**) are present in the dataset, improving consistency and accuracy during analysis.

Q2 : Covariance and Data Reduction in E-commerce Analytics

1. Introduction

The e-commerce customer dataset contains financial and behavioral attributes such as Age, Income, Purchase Value, Credit Score, Orders, Average Balance, EMI, and Customer Class. Before building a predictive model, it is necessary to analyze the relationships between attributes, remove redundant features, and standardize the data. These preprocessing techniques improve the quality of the dataset and enhance model performance.

A. Covariance Analysis

(i). Covariance Calculation

The relationship between Income and Purchase_Value was calculated using Microsoft Excel. The COVARIANCE.S() function was used to measure how the two variables vary together.

Excel Formula Used:

=COVARIANCE.S(C2:C9,D2:D9)

The screenshot shows a Microsoft Excel spreadsheet with a dataset of e-commerce customer information. The data is organized in columns: Cust_ID, Age, Income, Purchase Value, Credit Score, Orders, Avg. Balance, EMI, and Customer Class. The formula bar at the top displays the formula =COVARIANCE.S(C2:C9,D2:D9), which is used to calculate the covariance between the Income and Purchase Value columns. The spreadsheet interface includes the standard ribbon (File, Home, Insert, Draw, Page Layout, Formulas, Data, Review, View, Help) and a status bar at the bottom.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
	Cust_ID	Age	Income	Purchase Value	Credit Score	Orders	Avg. Balance	EMI	Customer Class														
1	U1	24	40	100	650	15	12	5000	Low														
2	U2	34	60	150	700	20	18	7000	Medium														
3	U3	44	85	210	720	28	25	9000	Medium														
4	U4	52	120	320	800	35	40	15000	High														
5	U5	29	45	110	680	18	14	5500	Low														
6	U6	39	75	180	710	24	22	8500	Medium														
7	U7	56	150	360	820	40	50	18000	High														
8	U8	37	70	160	705	22	20	8000	Medium														

The calculated covariance value is:

Covariance = 3529.464

	A	B	C	D	E	F	G	H	I
	Cust_ID	Age	Income	Purchase_Value	Credit_Score	Orders	Avg_Balance	EMI	Customer_Class
1	U1	24	40	100	650	15	12	5000	Low
2	U2	34	60	150	700	20	18	7000	Medium
3	U3	44	85	210	720	28	25	9000	Medium
4	U4	52	120	320	800	35	40	15000	High
5	U5	29	45	110	680	18	14	5500	Low
6	U6	39	75	180	710	24	22	8500	Medium
7	U7	56	150	360	820	40	50	18000	High
8	U8	37	70	160	705	22	20	8000	Medium

(ii). Interpretation of Covariance

The covariance value obtained is 3529.464, which is a positive value. This indicates that Income and Purchase_Value have a positive relationship. Customers with higher income generally tend to spend more on purchases, showing that both attributes increase together.

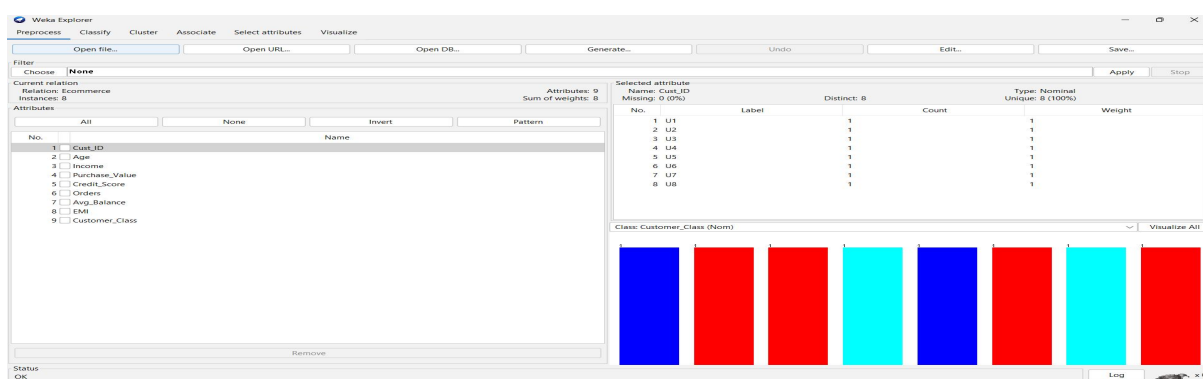
(iii). Importance of Covariance

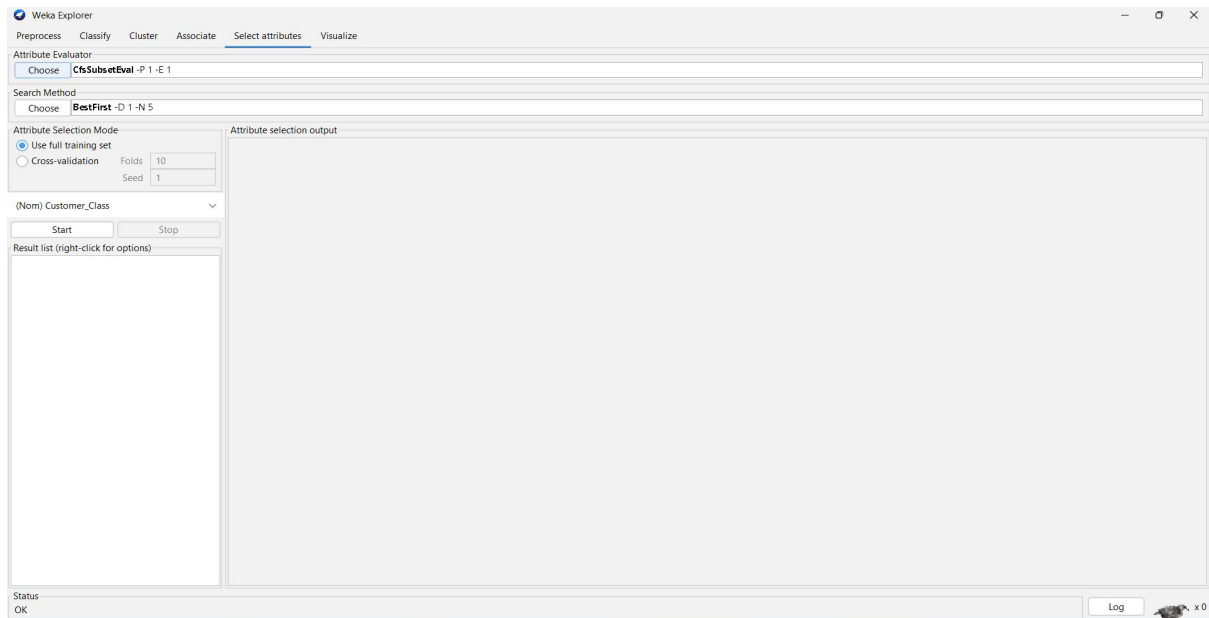
Covariance is a statistical measure used to determine the relationship between two numerical variables. It helps identify whether two attributes move together in the same direction or in opposite directions. This information is useful during feature selection because highly related attributes can be analyzed before building predictive models.

B. Feature Selection

(i). Identification of Important Features

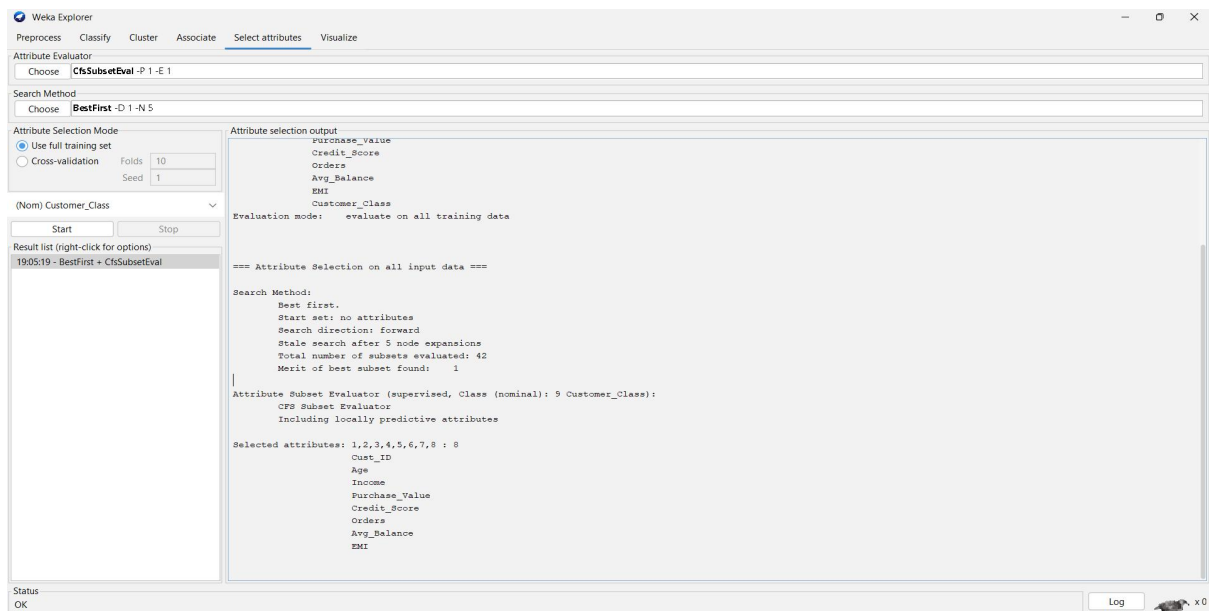
The dataset was imported into WEKA and the Select Attributes panel was used for feature selection. The CfsSubsetEval evaluator and BestFirst search method were selected to identify the most relevant attributes for classification.





(ii). Selected Features

After executing the feature selection process, WEKA selected the most informative attributes from the dataset. These selected features contribute significantly to customer classification, while redundant attributes can be removed to simplify the model.



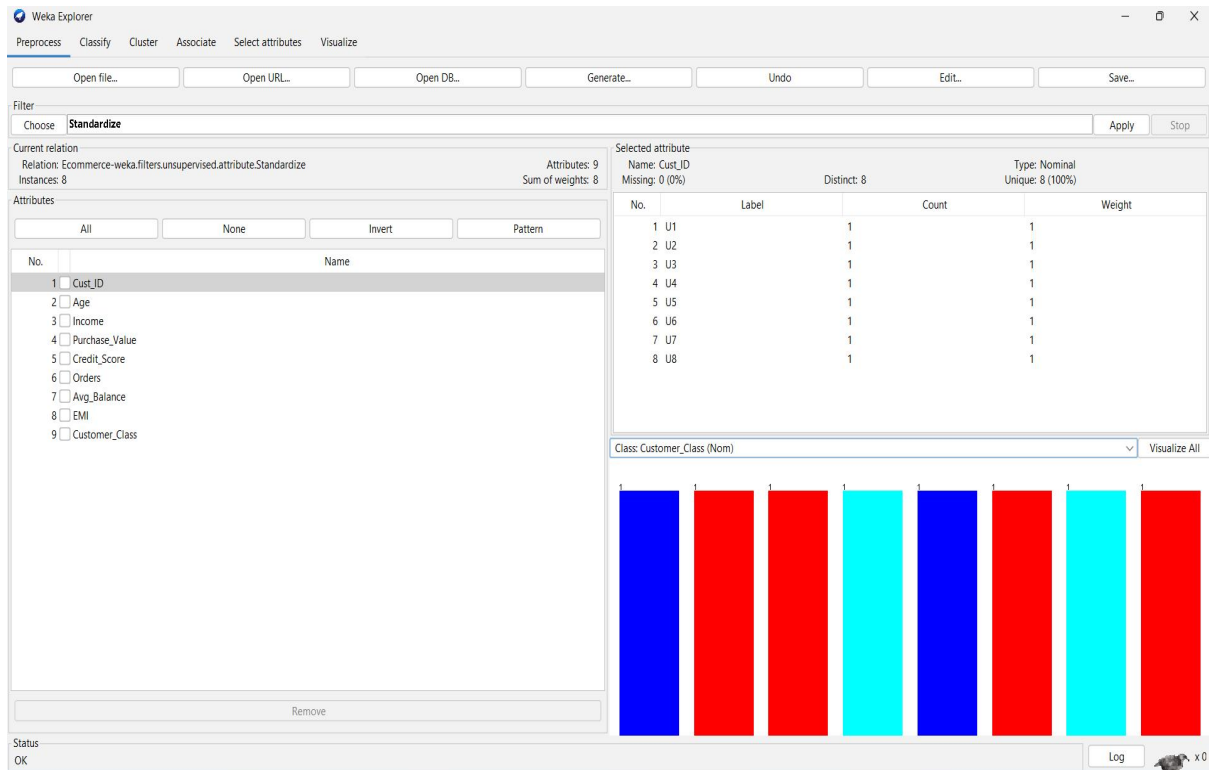
(iii). Justification

Feature selection reduces the number of unnecessary attributes in the dataset. It decreases computational complexity, improves model accuracy, and helps prevent overfitting by retaining only the most relevant features.

C. Data Transformation

(i). Standardization of Numerical Attributes

The Standardize filter available in the Preprocess tab of WEKA was applied to the numerical attributes of the dataset. During this process, identifier and class attributes such as Cust_ID and Customer_Class were excluded from transformation.



(ii). Need for Standardization

Standardization transforms numerical values to a common scale by adjusting their mean and standard deviation. This prevents attributes with larger values from dominating the learning process and improves the efficiency of machine learning algorithms that are sensitive to differences in data scale.

2. Discussion

The dataset was successfully analyzed using covariance analysis, feature selection, and data transformation techniques. Covariance confirmed the positive relationship between Income and Purchase Value. Feature selection identified the most significant attributes required for classification, and standardization transformed the numerical data into a uniform scale. These preprocessing steps improve data quality and prepare the dataset for effective predictive modeling.

Q3 : Data Transformation & Discretization in Education Analytics

1. Introduction

The education dataset contains student details such as Study Hours, Test Score, Attendance, and Performance. Since Study Hours and Test Score are continuous numerical attributes, they can be transformed into meaningful categories using discretization techniques. This improves data interpretation and supports better decision-making in educational analytics.

A. Equal-Width Binning

(i). Applying 3 Bins for Study_Hours and Test_Score Study_Hours

Minimum Value = 2

Maximum Value = 11

Range = $11 - 2 = 9$

Number of Bins = 3

Bin Width = $9 \div 3 = 3$

Bin Intervals

Bin	Interval
Bin 1	2 – 5
Bin 2	5 – 8
Bin 3	8 – 11

Study_Hours after Equal-Width Binning

Student	Study_Hours	Bin
S1	2	Bin 1
S2	3	Bin 1
S3	4	Bin 1
S4	5	Bin 1
S5	6	Bin 2
S6	7	Bin 2
S7	8	Bin 2

S8	9	Bin 3
S9	10	Bin 3
S10	11	Bin 3

Test_Score

Minimum Value = 40

Maximum Value = 100

Range = $100 - 40 = 60$

Number of Bins = 3

Bin Width = $60 \div 3 = 20$

Bin Intervals

Bin	Interval
Bin 1	40 – 60
Bin 2	61 – 80
Bin 3	81 – 100

Test_Score after Equal-Width Binning

Student	Study_Hours	Bin
S1	40	Bin 1
S2	50	Bin 1
S3	55	Bin 1
S4	60	Bin 2
S5	75	Bin 2
S6	85	Bin 3
S7	90	Bin 3
S8	95	Bin 3
S9	98	Bin 3
S10	100	Bin 3

B. Equal-Frequency Binning

The dataset contains 10 records. These records are divided into 3 bins, with each bin containing approximately the same number of records.

Study_Hours Equal-Frequency Bins

Bin	Values
Bin 1	2, 3, 4
Bin 2	5, 6, 7
Bin 3	8, 9, 10, 11

Test_Score Equal-Frequency Bins

Bin	Values
Bin 1	40, 50, 55
Bin 2	65, 75, 85
Bin 3	90, 95, 98, 100

C. Concept Hierarchy Construction

Study_Hours Hierarchy

Study_Hours	Category
2 – 4	Low
5 – 7	Medium
8 – 11	High

Test_Score Hierarchy

Test_Score	Category
40 – 60	Poor
61 – 80	Average
81 – 100	Excellent

These concept hierarchies simplify numerical data into meaningful categories, making it easier to analyze student performance.

Q4 : Normalization Techniques

1. Introduction

Normalization is a preprocessing technique used to transform numerical data into a common scale without changing the relationships among the values. It improves the performance of machine learning algorithms by ensuring that attributes with larger values do not dominate those with smaller values.

A. Min-Max Normalization

Given Data

200, 300, 400, 600, 1000

Formula

$$\text{Normalized Value} = \frac{X - \text{Minimum}}{\text{Maximum} - \text{Minimum}}$$

Minimum = 200

Maximum = 1000

Range = 800

Value	Calculation	Normalized Value
200	$(200-200)/800$	0.000
300	$(300-200)/800$	0.125
400	$(400-200)/800$	0.250
600	$(600-200)/800$	0.500
1000	$(1000-200)/800$	1.000

B. Z-Score Normalization

Formula

$$Z = \frac{X - \mu}{\sigma}$$

Where:

- μ (Mean) = 500
- σ (Standard Deviation) $\approx$ 316.23

Calculations

Value	Z - Score
200	-0.95
300	-0.63
400	-0.32
600	0.32
1000	1.58

2. Conclusion

Both normalization methods successfully transform the data into a common scale. Min-Max Normalization converts values to the range 0 to 1, while Z-Score Normalization expresses each value based on its distance from the mean in terms of standard deviations.

These techniques improve data consistency and prepare datasets for effective data mining and machine learning applications.

Q5 : Data Transformation in Online Shopping Behavior Analysis

1. Introduction

The given dataset contains the visit duration of customers on an online shopping website. The time spent by customers varies from a few seconds to several minutes, resulting in a highly skewed dataset. To prepare the data for clustering and customer behavior analysis, data transformation techniques such as equal-frequency binning, Z-score normalization, and Min-Max normalization are applied. These techniques help organize the data, reduce the effect of varying scales, and improve the performance of data mining algorithms.

A. Equal-Frequency Binning

(i). Divide the Dataset into 3 Bins

The dataset contains 10 records.

Number of bins = 3

Each bin should contain approximately 3 to 4 records.

Visit Duration Data

15, 30, 45, 60, 120, 300, 600, 900, 1800, 3600

Equal-Frequency Bins

Bin	Visit Duration (seconds)
Bin 1	15, 30, 45
Bin 2	60, 120, 300
Bin 3	600, 900, 1800, 3600

(ii). Assign Each Customer to a Bin

Customer ID	Visit Duration (seconds)	Bin
C1	15	Bin 1
C2	30	Bin 1

C3	45	Bin 1
C4	60	Bin 2
C5	120	Bin 2
C6	300	Bin 2
C7	600	Bin 3
C8	900	Bin 3
C9	1800	Bin 3
C10	3600	Bin 3

(iii). Analysis

Equal-frequency binning divides the data into groups containing nearly the same number of records. The first bin represents customers with short visit durations, the second bin represents medium visit durations, and the third bin represents customers who spend a long time on the shopping website. This method simplifies the data and makes customer segmentation easier.

B. Z-Score Normalization

(i). Calculate Mean and Standard Deviation

Given Data

15, 30, 45, 60, 120, 300, 600, 900, 1800, 3600

Mean (μ)

$$\mu = \frac{15 + 30 + 45 + 60 + 120 + 300 + 600 + 900 + 1800 + 3600}{10} = 747$$

Mean = 747 seconds

Standard Deviation (σ) $\approx$ 1114.08

(ii). Apply Z-Score Normalization

Formula

$$Z = \frac{X - \mu}{\sigma}$$

Visit Duration	Z - Score
15	-0.66
30	-0.64
45	-0.63
60	-0.62
120	-0.56
300	-0.40
600	-0.13
900	0.14
1800	0.94
3600	2.56

(iii). Discussion

The Z-score method expresses each visit duration based on its distance from the mean. Extremely high values such as 1800 and 3600 seconds have larger positive Z-scores, making them easy to identify as extreme observations. This method is useful when comparing values with different scales.

C. Min-Max Normalization

(i). Apply Min-Max Normalization

Formula

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Minimum Value = 15

Maximum Value = 3600

Range = 3600 – 15 = 3585

(ii). Normalized Values

Visit Duration	Normalized Value
15	0.000
30	0.004
45	0.008
60	0.013
120	0.029
300	0.080
600	0.163
900	0.247
1800	0.498
3600	1.000

(iii). Comparison

Min-Max Normalization converts all values into the range 0 to 1, making the dataset suitable for machine learning algorithms that require scaled input values. In contrast, Z-score normalization transforms values according to their distance from the mean and highlights extreme observations. While Min-Max scaling is simple and easy to interpret, Z-score normalization is more effective for identifying outliers in the dataset.

2. Conclusion

In this experiment, the visit duration dataset was transformed using equal-frequency binning, Z-score normalization, and Min-Max normalization. Equal-frequency binning grouped customers based on similar visit durations, while Z-score normalization standardized the data around the mean and highlighted extreme values. Min-Max normalization rescaled all values between 0 and 1, making the dataset suitable for clustering and customer behavior analysis.

These preprocessing techniques improve data quality and support more effective online shopping analytics.